

ON THE SECURITY OF AERONAUTICAL DATALINK COMMUNICATIONS: PROBLEMS AND SOLUTIONS

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Abstract

Numerous protocols allow modern aircraft to communicate with ground entities over wireless networks, including the so-called Datalink digital communications protocols such as ACARS and FANS-1/A. Among other benefits, they greatly enhance automation and allow communication between embedded avionics and aircraft components with ground infrastructure. Unfortunately, none of these protocols incorporate any form of message authentication or confidentiality. To date, no security counter-measures have been proposed to address this with the exception of the ARINC 823 ACARS Message Security (AMS) standard currently employed by the US Air Force to communicate with the Federal Aviation Authority (FAA) air traffic controllers.

In this paper, we present a threat analysis of the security flaws in the context of modern usage Datalink communications in aviation. To do so, we first describe how Software Defined Radios (SDR) have made easy to mount impersonation and message spoofing attack on both ACARS and FANS1/A datalink protocols. We then evaluate the potential impact of such attacks on both aircraft safety and air traffic management. To lend credence to our analysis, we describe a proof-of-concept implementation of this attack with a Universal Software Radio Project (USRP) SDR. Finally, we studied the viability of widely adopting AMS as an authentication solution by analyzing its real-world impact in terms of frequency congestion. We show that the widespread adoption of AMS, or an equivalent solution, by all commercial aircrafts would be sustainable.

Introduction

Long gone are the days when air traffic controllers (ATC) used flags to guide pilots. Voice radio communications first replaced them, and now modern communication systems permit air-to-ground exchange of messages that ensure efficient ATC

operations and make air travel one of the safest modes of transportation.

Among these systems, Datalink communications allows pilots, controllers and airline operations to communicate with each other by sending data over radio waves. Datalink offer numerous advantages over classical voice communications and are now mandatory on many commercial air routes, having become a vital primary communication medium for most flights. Some Flight Management Systems (FMS) can interpret Datalink messages and automatically change flight parameters, which can then be executed by the autopilot with minimal pilot intervention.

These increased levels of integration and automation have greatly benefited the air transport industry by making it more efficient in its use of the airspace and even potentially safer. On the other hand, it has also increased the attack surface of aviation towards deliberate cyberattacks that target the Datalink communication systems and the avionics. This potential threat is exacerbated by the popularisation and commercialisation of Software Defined Radios (SDR) and related development frameworks. SDR could allow hackers to intercept and send spoofed Datalink traffic that appear to come from ATC or from airborne aircraft, which may seriously interfere with aircraft and air traffic control operations. SDR technology is available at a very low cost and has a low barrier of entry in terms of technical knowledge-. This would allow a vast number of motivated attackers to mount such attacks with relatively little risk of getting caught.

Older protocols such as the Aircraft Communication Addressing and Reporting System (ACARS) and Controller Pilot Datalink Communication (CPDLC) are particularly vulnerable to spoofing attacks, as security was not considered when they were designed. Indeed, they do not incorporate any cryptographic mechanisms for message confidentiality or integrity. Yet, they still are widely used and now offer a prime target for hackers.

In this paper, we first briefly describe the various Datalink communication protocols currently being used. We explain how they work and how they are being employed. We then conduct a threat analysis on several Datalink protocols. We describe how it is possible to take advantage of the lack of authentication in the ACARS and FANS1/A Datalink protocols in order to perform an impersonation and message spoofing attack. We complete our analysis by presenting different attack scenarios to currently flying airline pilots, which allows us to evaluate their potential impact on both efficiency of air traffic management and aircraft safety. In particular, we have successfully identified an attack scenario where ACARS messages containing take-off speeds obtained from weight & balance calculations, could be spoofed by an attacker, which in turn could have serious consequences leading to severe aviation incidents or accidents. To validate our hypothesis, we developed a proof-of-concept implementation of this attack with a Universal Software Radio Project (USRP) SDR using the Gnuradio framework. Finally, we use the ACARS traffic dataset gathered over a period of several weeks to study whether the adoption of ARINC 823, a secured version of ACARS used by the US Air Force to interact with civilian controllers, could be viable from a technical point of view, in particular with respect to bandwidth usage overhead.

Datalink Landscape

Uses

“Datalink” is a collective term used to refer to numerous aeronautical communication protocols which allow digital data exchanges—as opposed to classical voice exchanges—over wireless networks. After its standardization in 1978, ACARS became the first operational datalink communication protocol. It was proposed by the private corporation ARINC, responsible for numerous aeronautical communication standards [1]. Datalink was quickly adopted for Air Traffic Control (ATC) and Aeronautical Operational Control (AOC) purposes, since it offers many benefits over voice exchanges. It reduces the possibility of misinterpretation messages, since they are displayed as text, – and frees congested VHF/HF radio frequencies due to a better bandwidth

management in comparison to voice communications.

The increased utilisation of ACARS led the ICAO to push forward recommendations in the form of a new standards, the Future Air Navigation System (FANS). To support these new technological needs, new VHF Data Links (VDL) networks appeared between 1991 and 1996 and serve as a physical layer to support them. Boeing and Airbus each developed their own implementation of the ICAO’s FANS concept, known as FANS-1 and FANS-A, respectively, which is why these implementations are often referred as FANS-1/A.. FANS-1/A communications are mostly used for clearance delivery and ATC operations when aircrafts are *En Route*, as response time from both controllers and pilots is not critical for flight safety [2].

Thanks to these new systems, avionics components can now directly communicate with the ground, thus highly enhancing automation. Some protocols such as CPDLC allow control instructions to be directly executed by the Flight Management System (FMS). It is worth mentioning that the ICAO, in collaboration with service providers such as ARINC and SITA, aims to develop an efficient and unique network for both AOC & ATC communication called *Air Traffic Network* (ATN). It is intended to replace present networks by 2025, but its exact specifications are yet to be fully defined [3]. The main Datalink protocols and their features are shown in Table 1.

In the rest of this paper, we will focus on ACARS and CPDLC communications as they are used for text transmission in safety-critical situations and present several important flaws.

State of the Art in Datalink security

Security Objectives

The 9/11 events triggered massive reactions within the aviation community leading to massive improvements in both safety and security. However, it is known that flaws still remains in aeronautical communication protocols, especially for those developed prior to this event when security was not such a key consideration [7], [8]. When it comes to the security of information systems, three main objectives must be considered: *availability*, *confidentiality*, and *authentication/integrity* [9].

Within the context of Datalink communications, these objectives raise very specific concerns.

Table 1. Datalink Protocols and Performances [4]–[6]

	<i>AOC</i>	<i>ATC</i>	<i>Technical specification</i>
ACARS	• Plain text communication	• Plain text communication • Departure clearances DCL • Oceanic clearances OCL • ATIS information D-ATIS	• Physical layer: VHF/HF/Satcom • Debit: 2.4 kbit/s • Modulation: MSK • Character oriented protocol
FANS-1/A FANS-2	• Not used for AOC	• ADS-C • CPDLC • ACARS over FANS-1/A	• Physical layer: VHF - on VDL sub-networks/HF/Satcom • Debit: 31.5 kbit/s • Modulation: D8PSK • Bit oriented protocol
ATN (future)	• IP communication	• ATC communications • CPDLC over ATN • AMC Link	• Yet to be fully operational & standardized

Availability is critical for ATC and AOC communication as they ensure adequate preparation and a smooth flight. Any disturbance within these systems would force the crew to communicate on auxiliary and less efficient networks. A Denial-of-Service (DoS) attack would leave the pilot without clearance instructions which could quickly result in serious disruptions in air traffic. Even if these communications are always redundant [10], attackers could simultaneously jam several frequencies. Availability is not only endangered by malicious attacks. Indeed, a lot of pilots already faced “stuck mike” situations, where one aircraft broadcasts continuously on a frequency, thus preventing others from communicating.

It is legitimate to want to maintain the *confidentiality* of datalink communications as they may contain sensitive information that could harm a company reputation or even the safety of flight.

However, the physical layer used for datalink communications is by nature easy to eavesdrop on and *confidentiality* can only be achieved through the use of cryptography. It is unclear to what extent airlines companies and National Aviation Authorities (NAA) want to protect the confidentiality of datalink exchanges. Consequently, neither ACARS nor FANS1/A employ any kind of cryptographic algorithms. This has allowed the emergence of a very active community of enthusiastic amateur and hobbyists that regularly intercepts, decodes and makes public the content of captured datalink messages [11], [12].

Authentication and integrity is achieved when every user of the system has absolute certainty that he has access to legitimate information. In aviation, Trust between actors and trust in the information they exchange is key to ensuring the proper conduct of aeronautical activities. For instance, a pilot should always be sure that he is talking to a real controller before he follows any control instruction. Attacks where the attacker poses as a legitimate user are called *impersonation attacks*. They have already been identified as serious threats by the ICAO and adequate measures to prevent them must be taken [13], [14]. The Melbourne airport already faced this kind of problem, when an unauthorised transmission gave false information to pilots through VHF radio [15]. As we will explain in Section 3, a similar attack on Datalink communications would be completely feasible since neither ACARS nor CPDLC incorporate a robust authentication scheme, which could have even worse consequences than spoofed VHF communications. As these flaws are now common knowledge, some efforts sustained by private companies have been made to address them.

Secured Datalink Protocols

The first efforts to protect Datalink communication came from the US Air Force. Military aircraft need to use ACARS and CPDLC communications as they are mandatory on most of the civil air traffic routes [16]. Because the Air Force was not willing to have information about their aircraft be easily available, a version of ACARS ensuring *confidentiality* and *authenticity* was needed. Joint work with Honeywell led to the development of a secure version of ACARS [17], and later the adoption of the ARINC 823 standard, ACARS Message Security (AMS). AMS is implemented at the application level thus resulting in easier

implementation inside aircraft avionics and allows perfect backward compatibility.

In AMS, *symmetric* cryptography was favoured as the main tool to ensure both confidentiality and authenticity due to efficiency. Confidentiality is achieved by having both parties (transmitter and sender) pre-share the same shared secret key *K*. In addition, this same key can be used to generate a Message Authentication Code (MAC) appended at the end of the original message. A MAC for a message *m* can be computed thanks to the shared secret *K* and a hash function *h*:

$$\text{MAC}_K(m) = h(K \parallel m) \tag{1}$$

However, such MAC are more often computed using an improved and more secure version of the Equation 1, normally the HMAC transformation described in the RFC 2104 standard.

Since it would not be practical for every airplane to pre-exchange a shared secret key with every air traffic control station with which it will communicate, AMS uses *asymmetric* cryptography to resolve the key distribution problem. In particular, ARINC 823 part 2 [19] prescribes the use of the UIT

X.509 standard for Public Key Infrastructure (PKI), the same as that used for securing SSL/TLS communications on the Internet, to achieve this aim.

PKI's rely on the notion of *chain of trust* between *certificate authorities* (CA) and the trust of users into those CA. At the top of this hierarchy of trust, are the root CA, that everyone recognizes and trust. These root CA can designate other CA that they consider trustworthy, who will in turn appoint others CA, etc. At the end of the certification hierarchy, authority closest to the users will directly certify their public keys. In aviation, the users would include aircraft, controllers and airline operations, who would only grant their trust in another user's public key if it is associated with a valid certificate, i.e. one for which the chain of trust can be followed all the way to a root CA.

Thus, AMS relies on both symmetric and asymmetric cryptography in order to achieve an acceptable level of security, while optimizing the use of available resources in term of network capacity and computation power. The cryptographic methods used in AMS are slightly more complex than the one we have just described, as is shown in Figure 1.

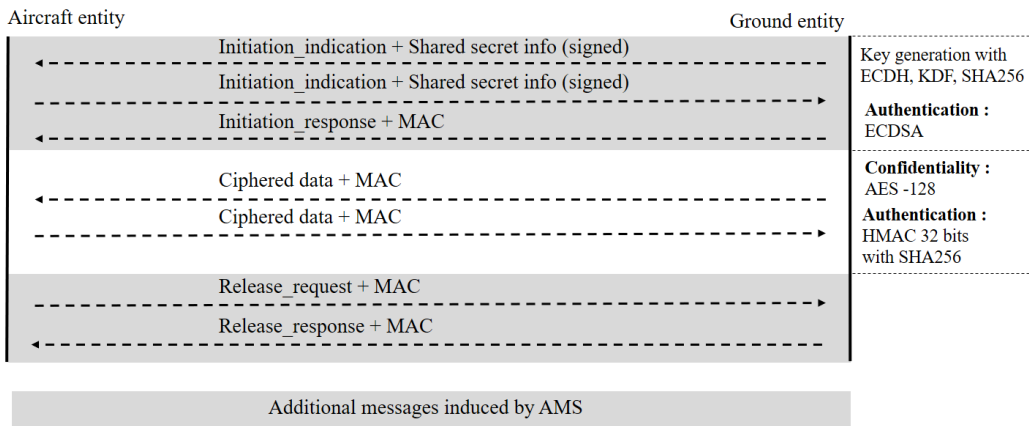


Figure 1. AMS Overview [18]

When a secure session is needed, shared secret keys are computed during an initial handshake. First, communicating parties define which cryptographic algorithms will be used. In AMS, the handshake is based on the Elliptic-Curve Diffie-Hellman (ECDH) algorithm. Once a secret shared key has been generated by both parties, confidentiality is achieved by using that key with the AES-128 algorithm, while authentication achieved by appending 32 bits from

the HMAC of the original message at the end of each such message. In session-less communication, i.e. sessions without shared secret to encrypt the messages, packets are signed using the Elliptic Curve Digital Signature Algorithm (ECSDA); however, these messages are not confidential. Therefore, session-less exchanges are only used in AMS for the three packets in the handshake.

There is currently no standard available providing the foundations for a secure version of CPDLC. However, the constraints are substantially the same (low bandwidth, wireless environment, low computation power, etc.) and numerous work suggests that a solution similar to AMS will be adopted for an upcoming version of secure CPDLC [8], [20].

It is both time-consuming and expensive to implement modifications in the avionics component software. Safety is only achieved thanks to demanding certifications and significant amounts of testing, thus greatly slowing down the roll-out of any modification in these systems. It is not economically viable - for manufacturers to implement security updates in their systems, in the same way as it is done in traditional Information Technology (IT) security practice. New iterations of the Datalink protocols have always come from private initiatives in order to meet specific market needs. Secure alternatives will not be put forward as long as it is believed that there is no need to change a system that has stood the test of time in terms of reliability. However, these beliefs have become hazardous given the recent advent of new amateur radio techniques and technologies such as those based on *Software Defined Radios* (SDR), with which it has never been easier to conduct attacks on Datalink systems.

Threat Analysis

There is always a gap between the security expectations of a system and the real world. Accordingly, we decided to conduct a rigorous threat analysis to evaluate the security risks associated with ACARS and CPDLC.

Methodology

A common approach to threat analysis consists in distinguishing threat actors and the vulnerabilities they could use in a particular attack scenario. Once this is done, it is necessary to evaluate the *expected impact* of exploitation of the identified flaw by an attacker. It is assessed by multiplying the *impact* of the attack by its *probability*.

The *probability* is obtained by averaging the coefficients assigned to:

- *Capacity*, which quantifies the intellectual and material resources needed to achieve an attack.
- *Motivation*, which represents the interest that an attacker would have to compromise a system.
- *Opportunity*, which evaluates the ease with which an attacker could exploit a flaw.

Threat Agents

A clear separation must be drawn between internal actors (pilots, controllers, airlines) and external actors. In order to obtain significant results, we must put aside the former as threat actors, as monitoring their actions would require very specific and complex methods, which are out of the scope of this study. We will therefore consider that every internal actor is benevolent and will not interfere with the on-going operations of flights and air traffic control.

It is also difficult to obtain a comprehensive list of all the potential external attackers of Datalink systems. However, we will consider the results obtained by recent works [21], [22]. Rogue states and terrorist groups are for instance prime suspects in an upcoming active attack against Datalink systems.

Classification of Security Objectives

We already described to what extent *confidentiality*, *availability*, and *authentication/integrity* are critical for flight safety. Furthermore; ICAO has already identified several important threats to datalink communications and wants to address them in its next generation air traffic control systems [13], [14]. We used this information to weight *capacity* *motivation* and *opportunity* on a scale of 4. The higher the score the more critical this aspect of the attack is. The results are presented in Table 2.

Table 2. Classification of Security Objectives

	Capacity	Opportunity	Motivation	Probability	Impact
Confidentiality	1	3	3	2.33	1
Availability	2	2	3	2.33	4
Authentication/ Integrity	4	3	3	3.33	4

Attack Scenarios

With regards to the lack of authentication in ACARS and CPDLC protocols, a lot of different attack scenarios can be considered. The most affected security objective is of course *authentication*. We

focus on impersonation attacks because they were already identified as highly critical by the ICAO. To study the viability and potential impact of these attacks, we collected ACARS and CPDLC communications within the Montreal Terminal Area by using an SDR in combination with the open-source decoder acarsdec [12]. This data set contains 123,409 ACARS and CPDLC messages collected intermittently during the period from 2017-12-07 to 2018-01-03, i.e 4407,5 message each day.

ACARS

Some clearances are still delivered via ACARS messages when the supporting communications network is not FANS1/A compliant. However, its main use is, and will probably remain, for AOC communications. We thereby focused on an attack involving communications between pilots and company operations. Figure 2 displays a real message received by an aircraft, whose content we partially redacted for legal and security reasons.

```
Freq: 131.550MHz
ACARS mode: 2 (50), label: C1 (Uplink message to the cockpit printer 1)
Block id: L (76)
Aircraft reg: [REDACTED]
Type: [REDACTED]
Airlines: [REDACTED]

Message content:-
[REDACTED]
AGM
[REDACTED]
- AERODATA PERFORMANCE UPLINK
-----
TAKEOFF DATA FLIGHT :8030 REL 1
[REDACTED] 2017 [REDACTED]
WIND 170/14
TEMP P15
QNH 30.00
GTOW 45567
INDEX
-----
REMARKS WET RUNWAY
WB DATA MISSING
-----
YUL 24L 9600 FT
DT H237 FLAP 8
FRA 1118 VI 137
FULL - RLNG - BL OPEN VR 137

N1 90.2 AT/--- V2 147
MAXN1 92.3 VT 175
MTOW 50143/A
-----
171 PROCESSED
```

Figure 2. AOC Message for V-Speeds Calculation

As can be seen, the message contains the optimal take-off speeds computed by the airline operations center, thanks to numerous parameters such as aircraft weight, runway length, Automatic

Terminal Information Service (ATIS) information. This message is normally sent to the pilots right before the push-back. According to standard definitions:

- V_1 represents the *critical engine failure recognition speed*, after which the take-off cannot be safely aborted.
- V_r represents the *rotation speed*, after which the pilot lifts the nose of the aircraft to initiate its rotations. If the speed, hence the lift, is sufficient the plane should leave the ground seconds after the beginning of the rotation.
- V_2 represents the *take-off safety speed*, after which it can start its ascension securely. [23]

The information necessary for the computation of these V-speeds can be obtained thanks to voice communication or ATIS. Once they are gathered, the pilot sends them to its airline operations centre, which will respond with in an unauthenticated ACARS message such as the one displayed in Figure 2.

The precise procedure depends entirely on the company as AOC protocols are not standardized and were often developed within the airlines. Some of them do not use ACARS to compute or exchange the V-speeds. However, those that rely on unauthenticated communications without any form of sequence control are at risk, as shown in Figure 3. This situation would allow attackers to spoof the original message with data used in the calculations or the latter message containing the results. After the analysis of one day's worth of transmissions, we already were able to reverse engineer and understand the complete structure of AOC communications on that topic for at least one airline company. An impersonation attack with the introduction of false data at any step in the process would result in modified V-Speeds being entered into the FMS or being considered by the pilots. This could result in aviation incident or accident, if the modifications were carefully chosen by the attackers. This is not far-fetched as there is at least one known instance of a tail strike accident occurring when pilots entered the wrong data to compute take off speeds, resulting in a botched take off [24]. The technical feasibility of this attack is demonstrated in the next part.

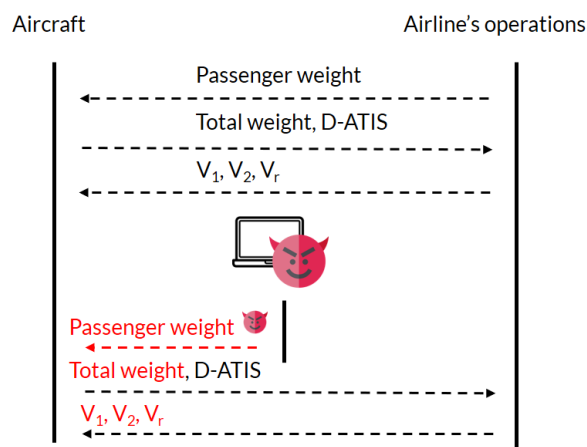


Figure 3. Tail Strike-Causing ACARS Exchange

We verified its credibility by interviewing several pilots from different airlines. Numerous parameters still have to be considered. For instance, the V-speeds are most of the time manually entered in the FMS. But targeting pilots when their attention is low, at the end of their working day for instance, could greatly increase the chances of success as they do not necessarily verify the data received via Datalink and could result in the situation presented in Figure 4. Furthermore, in order to meet the increasing demand for automation, some manufacturers such as Airbus now offer systems which allow the FMS to directly read information received via ACARS and bypass pilot action [25].

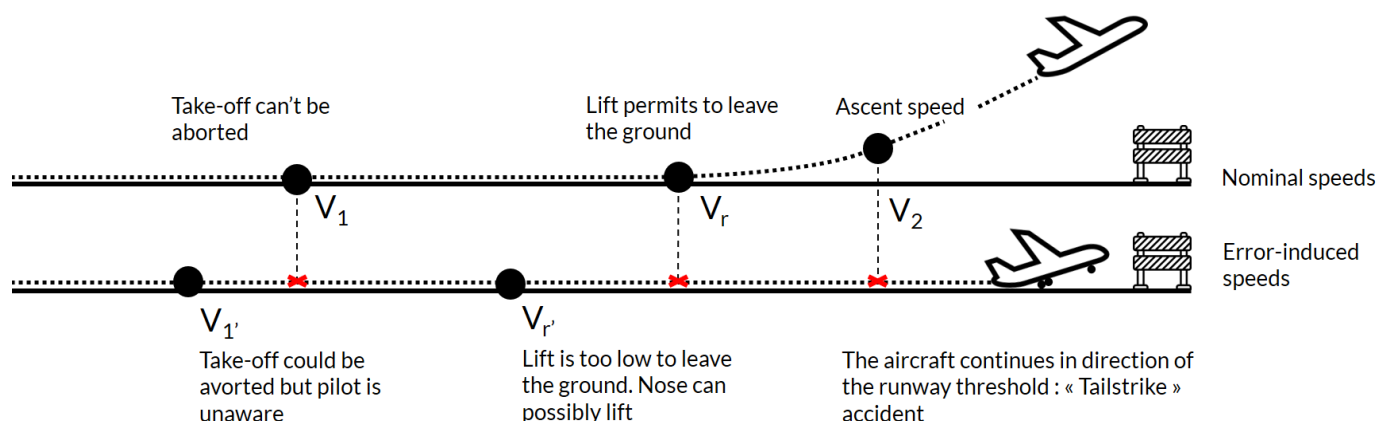


Figure 4. ACARS-Caused Tail Strike Scenario

CPDLC

As it was directly designed to transmit air traffic control information, it is possible to devise worrisome attack scenarios on CPDLC. While some instructions are still transmitted via ACARS or ACARS over FANS1/A, most of them are sent via unauthenticated CPDLC packets when over the ocean.

It is now possible for an attacker to impersonate an air traffic controller, with all the possible implication which this entails. An attacker could order a plane to follow any path modification and chances are that the instructions would be followed. Indeed, there is no reason for a crew to deny an imperative instruction which seems legitimate, especially since pilots greatly trust their systems [26]. Since standard procedures between controllers and pilots include a lot of instructions—as described by

the ICAO [10]—it leaves great room of manoeuvre for attacks. Therefore, we will not precisely describe every instruction set which could lead to disruption in air.

It is worth noticing that pilots have already have to deal with unintentional erroneous instructions due to bogus CPDLC messages. A “DESCEND AT * MAXIMUM” was once received by an aircraft, whose pilot fortunately asked for voice confirmation [27]. However, some crews do not pay close attention to the origin of the instructions they receive and have followed instructions targeted at previous flights [28]. It would be possible for an attacker to disrupt air traffic in a vast area if he were to send appropriate messages to specific aircraft.

Aviation communications and exchanges are highly procedural. A plane deviating from its

nominal route without any controller action would be quickly identified. However, it is unclear how fast the existing procedures would trigger a reaction. Thus, the interval of time an attacker would have to induce a collision before he would be detected has yet to be assessed.

Experimental Proof of Concept

We described two plausible attack scenarios on ACARS and CPDLC communication. They both exploit the lack of authentication in Datalink communications. This flaw has long been known and identified as most, worrisome [13]. Nonetheless, those security problems have become easily exploitable because of the democratisation and commercialisation of SDR and associated frameworks. These tools allow easy reverse engineering of communication protocols, including aeronautical ones. They significantly decrease the knowledge and material resources necessary to intercept, and interfere with legitimate Datalink communications, thus allowing more a larger population of potential attackers to target aviation communication protocols.

We developed a proof of concept attack with a Universal Software Defined Radio (USRP B200) from Ettus research and the *Gnuradio* open source framework. Such programmable hardware can be purchased online for a few hundred dollars and can be programmed with a simple laptop via a USB port. We focused on ACARS as it is a relatively simple

protocol, but a similar approach could be adopted for CPDLC communications.

We had access to several standards such as ARINC-618 and ARINC-823, which considerably expedited our work, but every necessary information is already available online.

The Minimum Shift Keying (MSK) modulation scheme used in ACARS messages can be described as follow:

- A low pitch (1200 Hz) is sent when the transmitted bit differs from the last sent bit.
- A high pitch (2400 Hz) is sent when there is no bit change.
- Transition between symbols should have no phase discontinuity and occur when tone amplitude is null.
- The amplitude of the VHF carrier should then be modulated with that baseband signal.

Figure 5 shows our implementation of the ACARS modulation scheme using only basic GNU radio blocks. We first XOR the bit stream of the outgoing message with a one-bit delayed version of itself. The resulting bit stream is then interpolated to obtain the desired sample rate and converted to symbols through a frequency modulation block where constants are chosen so that a “1” increases the phase by π radian and a “0” by 2π radian.

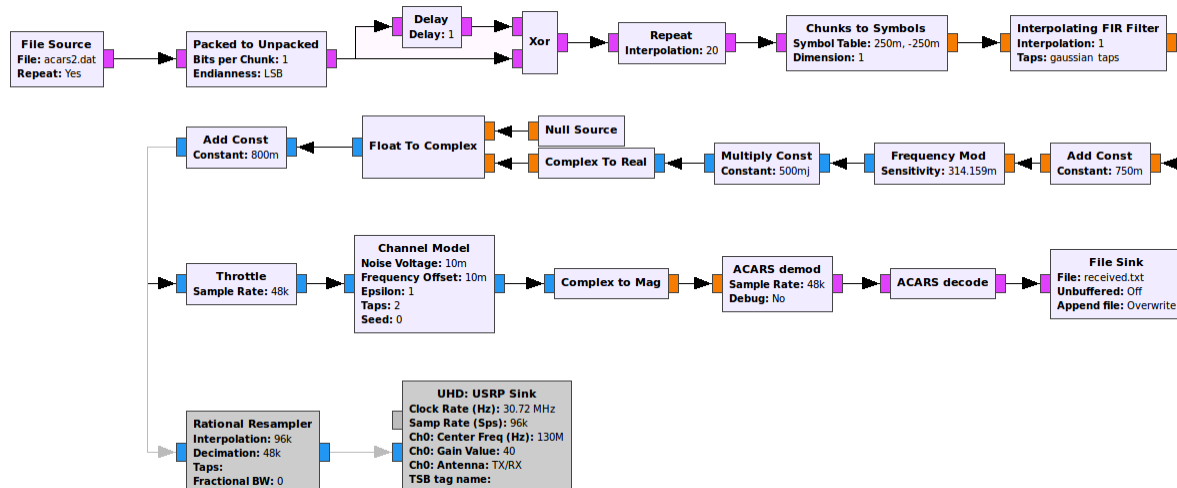


Figure 5. Gnuradio Implementation of ACARS TX/RX

Therefore, the real part of the resulting signal describes a full sine wave when two consecutive bits of the outgoing message are identical, and half a sine wave when they differ. A gaussian filter is also used to reduce out-of-band interference by limiting sideband power. At this point the signal has a modulus of 1 which is of course not suitable for amplitude modulation. Hence the imaginary part is cancelled out in the remaining blocks. The two last constants determine the modulation index.

```
Found Rafael Micro R820T tuner
Exact sample rate is: 2500000.107620 Hz

[#1 (F:131.550 L: -9 E:0) 09/03/2018 05:21:39 -----
Mode : 2 Label : H2 Id : 2 Nak
Aircraft reg: .IC-NSE Flight id: AA-000
No: 0001
HELLO-WORLD-POC-ATTACK-ON-ACARS
```

Figure 6. False Packets on AcarsDec

Datalink Network Resilience

Impact of ARINC 823 on the Network

The ARINC 823 standard - ACARS Message Security (AMS), is the secured version of ACARS adopted by the US military. Since it implements a strong authentication mechanism, it could effectively protect against this type of attack. Accordingly, Civil aviation should ideally adopt a similar solution as quickly as possible to avoid serious consequences derived from such attacks. However, it is unclear whether the VHF/HF and satcom networks would be able to withstand the increased traffic induced by secure session initialisation and authentication measures, i.e. MAC and digital signatures.

Cryptographic measures implemented in AMS can cause an increase in the average ACARS traffic for two main reasons:

1. Although compression algorithms are applied to the payload, both authentication techniques—HMAC and ECDSA—require a dedicated field inserted in the original message. This overhead can exceed the size limit of an ACARS packet, i.e. 210

We tested our implementation by simulating ACARS transmissions in GNU Radio. We used the gr-acars2 software for testing the reception and a Channel Model block for simulating the channel perturbations. Real transmissions were also tested between a USRP and the acarsdec receiver in a laboratory setting in order to avoid transmission on aviation frequencies. Figure 6 shows the results of such an ACAS message transmission.

characters in downlink and 220 in uplink. If the size of the authenticated message exceeds this limit, it is split into up to eighteen data frames, causing the appearance of new messages on the network.

2. AMS offers confidentiality thanks to dedicated secure sessions. At least five new messages, described in Figure 1, are needed to initiate and terminate these sessions.

We intercepted and decoded ACARS messages at Montreal Trudeau airport for twenty-eight days on the frequency 131.550 Mhz (the principal worldwide ACARS frequency used for AOC & control communications). This allowed us to have a good insight of the impact that the generalisation of the use of AMS on the datalink traffic. The result we obtained are displayed in Table 3 and the boxplot associated with packet length is shown in Figure 7. Intercepted packets and python scripts used to process them are available on demand. An average number of 92.25 unique planes transited each day over Montreal throughout our observation period. The establishment and completion of secure sessions would have triggered the transmission of 461.25 additional messages per day, representing a **10.5%** increase.

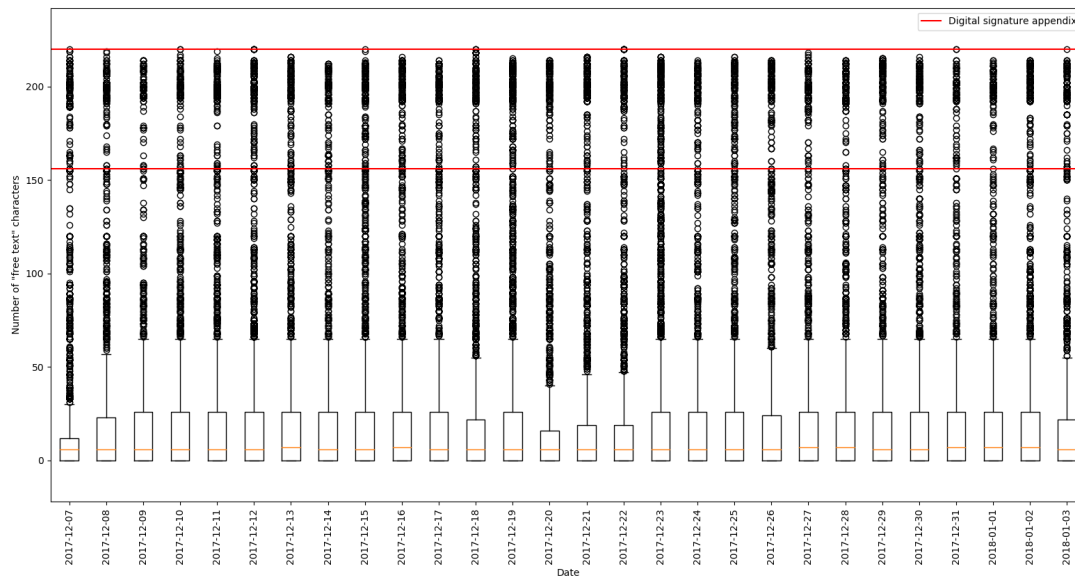


Figure 7. Boxplot of Received ACARS Message Length

Table 3. ACARS Traffic Increase Sources

Increase source	Additional messages	Traffic increase
32-bit HMAC	20	0.016%
64-bit HMAC	256	0.21%
128-bit HMAC	1671	1.35%
Handshake	12915	10.5%

It is difficult to assess the effective traffic load on the network as every airport and every route presents a different traffic and specific control procedures. They may rely on datalink communication or be fully operated by voice. However, the distribution of messages labels represents a good insight on Datalink operations in Canada. It also gives accurate information about the average size of ACARS messages. With this information, we can compute the approximate increase in bandwidth usage that the use of cryptography to secure the protocols would induce.

The “natural” growth of the global air traffic has been estimated between **1.6%** and **3.9%** each year [29]. Datalink communications should at least follow a similar development, and probably faster as aircraft board more and more avionics systems sending additional data over the network. A sudden increase

of 10% in datalink exchanges induced by cryptographic protections represents the equivalent of 2 to 6 years of normal evolution of the traffic. In sparsely populated regions like Quebec, there is no risk of congestion in the near future. However, in busy areas such as near important hubs, more frequencies will have to be used.

It is vital to promptly adopt protection measures, as the consequences of these increasingly likely attacks on Datalink systems could be disastrous. The solutions already exist and should be implemented quickly inside every commercial aircraft. As it stands, the increase in the network usage, however, could be a major constraint to the deployment of secure Datalink protocols. Therefore, we propose different solutions which could cut down the bandwidth overhead induced by cryptographic fields and messages.

Possible Optimisations of ARINC 823

The main source of increase in the number of exchanged packets are the handshakes necessary for the initialisation of a secure session between two actors. However, the interception and comprehension of the messages by an attacker alone does not harm air operations in any meaningful way. The main danger is, as we already detailed, in impersonation attacks. In other words, integrity is the most important security objective *not* confidentiality. We

therefore decided to abandon the confidentiality objective as authentication of communications alone should be sufficient to prevent active impersonation attacks. A badly authenticated message should simply be rejected by the avionics and will not be able to interfere with the aircraft system or the pilots.

In order to carry out an *authentication only* version of AMS, every packet should be signed with a digital signature. These signed packets are already used for the communication initialisation procedure of AMS when a 64-character ECDSA digital signature is appended to the message. This greatly reduces the payload size of ACARS messages to 146 characters in downlink and 156 in uplink. Slight modifications of AMS could lay the foundation for an effective and resource-efficient protection method:

A ***ground-to-air-only*** authentication mode could be used to authenticate only critical uplinks messages (e.g. label C1 for uplink messages to the cockpit printer or label H1 for terminal communication). This would be effective to avoid attacks on ATC communications as a pilot would always be certain of the legitimacy of received instructions. Certificates for every control centre crossed en route and emergency ones could be available in a database received by crews prior to their flights. We estimate that this would cause a **0.65%** increase in the ACARS traffic. This mode would still leave room for attacks such as the one we described above but would still greatly enhance the security while minimizing the impact on the network.

An ***all-authenticated*** mode where messages from aircraft could also be authenticated, could prevent attack scenarios such as those described above where *V*-speeds would be wrongly computed by the airline operations centre based on spoofed ACARS messages from the aircraft. In this case, it would not be practical for a database containing the certificate for every aircraft in the world to be created and distributed to all other aircraft or air traffic control authorities. An alternate solution is thus for each aircraft to send its certificate prior to entering a new area, in order to establish the authenticity of future messages. Authenticating every single ACARS packet via digital signature would result in a **6.61%** increase in traffic. Even if this would be less efficient than the ***ground-to-air-only*** mode, it would ensure a more resilient security protection against impersonating attacks.

These optimisations do present a major drawback: they require changes in the AMS standard, which does not allow authentication-only operations via digital signature. The path to their certification and large-scale use will therefore be tedious. However, we are convinced that this is a good start to implement cryptographic solutions for Datalink communications based on already available solutions. The deployment of AMS is not prohibitive in terms of bandwidth use as it causes at most the loss of a few years of traffic evolution. This may be not so critical in the grand scheme of things, considering that frequency usage is decades from reaching maximum capacity. Having to deploy replacement standards a few years earlier is a small price to pay to protect commercial aviation from such potentially devastating cyber attacks. Therefore, we very strongly recommend that a modified AMS or a similar solution be quickly developed and deployed as it would efficiently protect aircraft from the highly worrisome attacks we presented.

Conclusion

The situation regarding Datalink security is quite complex. Security standards exist but they are not used by airlines because they do not see them as a priority. However, the recent democratisation of SDR has completely turned the tables, and would-be attackers are now able to seriously disrupt air traffic by exploiting Datalink flaws.

We conducted a threat analysis that allowed us to discover credible attack scenarios. Active impersonation attacks could trigger major disruptions in air traffic control and global aeronautical operations. It could be in the form of *V*-speeds modification, false ATC instructions, or other scenarios. We have shown that these attacks must be taken seriously, as it is relatively easy to develop a proof-of-concept attack with scarce financial and intellectual resources and limited or no access to privileged information. The rise in bandwidth use induced by the cryptographic measures we presented may be compared with 2 to 6 years of nominal airline traffic increase. Therefore, we believe that they should be implemented as soon as possible. If datalink networks were to be already a few years from congestion, we showed that some optimisations are possible with slight modifications in already available standards such as ARINC 823 ACARS Message Security (AMS).

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